

| COURSE DESCRIPTION         |                                                                                                                     |               |     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|-----|
| Program                    | Electrical & Electronics Engineering, Electrical Engineering, Electrical Engineering & Electric Vehicles Technology |               |     |
| Course Title               | Basic Electrical and Electronics Engineering                                                                        |               |     |
| Course Code                | 1031                                                                                                                | Semester      | I   |
| Type of Course             | Practicum                                                                                                           | Contact Hours | 90  |
| Teaching Scheme (L: T:P:R) | 3:0:3:0                                                                                                             | Credits       | 4.5 |
| CIE Marks                  | 40                                                                                                                  | ESE Marks     | 100 |

### Introduction

This course provides a foundational introduction to basic electrical and electronic circuits. Students will learn key circuit parameters (such as voltage, current, resistance, and power) and how to measure them using standard instruments. The course also covers essential concepts of AC systems, along with electrical safety practices and common safety devices. Through hands-on activities, learners will explore passive components and develop practical skills in building circuits on a breadboard. Finally, the course introduces active components through applications such as rectifier circuits and basic logic gate configurations.

### Course Objectives

|   |                                                                                                                                |
|---|--------------------------------------------------------------------------------------------------------------------------------|
| 1 | To introduce fundamental concepts of electrical circuit parameters and their measurement techniques.                           |
| 2 | To impart knowledge about AC systems, electrical safety precautions, and safety devices                                        |
| 3 | To provide a functional understanding of passive electronic components and develop skills in constructing electronic circuits. |
| 4 | To provide knowledge about active electronic components in rectifier circuits and logic gate configurations.                   |

### Course Pre-requisite

| Topic | Course code | Course Title | Semester |
|-------|-------------|--------------|----------|
| NIL   |             |              |          |

### Course Outcome

| CO  | Course Outcome                                                                                                               | Blooms Taxonomy Level |
|-----|------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | Apply the basic laws and concepts to solve DC circuits and effectively use instruments to measure the electrical parameters. | Apply                 |
| CO2 | Construct an AC circuit to measure electrical parameters from the fundamental knowledge of AC systems and safety devices.    | Apply                 |
| CO3 | Apply the concepts of passive components to construct simple electronic circuits.                                            | Apply                 |
| CO4 | Apply the knowledge of electronic components for constructing the rectifier circuits and logic gate configurations.          | Apply                 |

### CO-PO mapping

| COURSE ARTICULATION MATRIX |                  |     |     |     |     |     |     |     |     |      |      |
|----------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| Course Outcomes            | PROGRAM OUTCOMES |     |     |     |     |     |     |     |     |      |      |
|                            | PO1              | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
| CO1                        | 3                | -   | -   | 3   | -   | -   | -   | -   | -   | -    | -    |
| CO2                        | 3                | -   | -   | 3   | -   | -   | -   | -   | -   | -    | -    |
| CO3                        | 3                | -   | -   | 3   | -   | -   | -   | -   | -   | -    | -    |
| CO4                        | 3                | -   | -   | 3   | -   | -   | -   | -   | -   | -    | -    |
| Total                      | 12               | -   | -   | 12  | -   | -   | -   | -   | -   | -    | -    |
| Correlation Level          | 3                | -   | -   | 3   | -   | -   | -   | -   | -   | -    | -    |

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-”

### Teaching and Assessment Scheme

| Teaching Scheme/Week                                                                                                                                                                                                                                                                                                                                       |   |   |   | Self-learning<br>(SS)/Week | Total<br>Hours/Semester | Credits<br>C | Examination Scheme |     |       |           |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|----------------------------|-------------------------|--------------|--------------------|-----|-------|-----------|-----|-------|
|                                                                                                                                                                                                                                                                                                                                                            |   |   |   |                            |                         |              | Theory             |     |       | Practical |     |       |
| L                                                                                                                                                                                                                                                                                                                                                          | T | P | S |                            |                         |              | CIE                | ESE | Total | CIE       | ESE | Total |
| 3                                                                                                                                                                                                                                                                                                                                                          | 0 | 3 | 0 | 6                          | 90                      | 4.5          | 20                 | 60  | 80    | 20        | 40  | 60    |
| <i>L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination</i> |   |   |   |                            |                         |              |                    |     |       |           |     |       |

### Syllabus - Major Topics

| Module     | Title                                                            | Major Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Instructional Hours |    |
|------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----|
|            |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | L                   | P  |
| Module I   | Fundamentals of Electrical systems and combinations of resistors | Basic terms in electricity - terms, definition, and units - current, voltage, resistance, power, and electrical energy.<br>Ohm's law - statement and simple problems<br>Series and parallel combinations of resistances, equivalent resistance (derivation and comparison), simple problems<br>Current division & voltage division rule, simple problems                                                                                                                                 | 11                  | 0  |
| Module I   | Fundamentals of Electrical systems and combinations of resistors | 1. Identify basic electrical circuit symbols<br>2. Differentiate functions of electrical measuring instruments<br>3. Use of the tester and multimeter<br>4. Demonstrate measurement of voltage and current in a DC circuit..                                                                                                                                                                                                                                                             | 0                   | 9  |
| Module II  | Basic parameters of AC systems and Electrical safety             | Basic parameters in AC systems: important terms-definition, units-cycle, frequency, time period, angular frequency, amplitude, instantaneous value, maximum value, rms value, average value, form factor, peak factor<br>Electrical safety- Safety precautions, Safety devices - fuse - miniature circuit breaker - earth leakage circuit breaker- RCCB<br>Power in AC circuits- types of power- active power, reactive power, apparent power, phase angle, and power factor, problems.  | 11                  | 0  |
| Module II  | Basic parameters of AC systems and Electrical safety             | 1. Electrical safety precautions<br>2. Electrical safety devices in residential installations<br>3. Measurement of power in an AC circuit                                                                                                                                                                                                                                                                                                                                                | 0                   | 9  |
| Module III | Different types of passive components                            | Resistors- types, applications, and colour coding of resistors<br>Inductance - definition-self inductance, mutual inductance, units, application of inductors.<br>Capacitance- definition, types- uses of capacitors, combination of capacitors, series and parallel combination                                                                                                                                                                                                         | 11                  | 0  |
| Module III | Different types of passive components                            | 1. Construct a simple electronic circuit on a breadboard.<br>2. Operate a DSO<br>3. Measurement of resistance, capacitance and inductance                                                                                                                                                                                                                                                                                                                                                | 0                   | 12 |
| Module IV  | Basic Electronic Components and Logic gates                      | Electronic components- active and passive components, types of semiconductors- intrinsic, extrinsic- doping- N type, P type<br>PN junction diode- symbol, construction, depletion region, working, biasing- forward and reverse biasing, applications, rectifier circuits-half-wave rectifier<br>Transistors- symbol, construction, working, types -NPN, PNP<br>Logic gates, definition, symbol, truth table of basic gates (AND, OR, NOT), universal gates(NAND, NOR), XOR, XNOR gates. | 12                  | 0  |
| Module IV  | Basic Electronic Components and Logic gates                      | 1. Identification of diode and transistor<br>2. Construction of a half-wave rectifier circuit<br>3. Verification of various logic gates                                                                                                                                                                                                                                                                                                                                                  | 0                   | 9  |

### Detailed Syllabus

| Subtopic         | Student Learning Outcome                                                          | Mode of delivery (L/T) | Mapped CO | Taxonomy Level | Instructional Hours |
|------------------|-----------------------------------------------------------------------------------|------------------------|-----------|----------------|---------------------|
| Module I         |                                                                                   |                        |           |                |                     |
| Basic parameters | Define current, voltage, resistance, power, and electrical energy and their units | L                      | CO1       | Remember       | 2                   |
| Ohm's law        | State Ohm's Law<br>Solve simple problems                                          | L                      | CO1       | Apply          | 3                   |

|                                             |                                                                                                                                                                                                                                                                                                                                                                             |   |     |            |   |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|------------|---|
| Resistance                                  | Define resistance<br>Derive equivalent resistance in series and parallel combinations.<br>Solve simple problems.                                                                                                                                                                                                                                                            | L | CO1 | Apply      | 3 |
| Current and Voltage Division Rule           | Explain the current division & voltage division rule.<br>Solve simple problems.                                                                                                                                                                                                                                                                                             | L | CO1 | Apply      | 3 |
| Electrical symbols                          | Identify basic electrical circuit symbols.                                                                                                                                                                                                                                                                                                                                  | p | CO1 | Remember   | 1 |
| Electrical measuring devices                | Differentiate the functions of a voltmeter, ammeter, wattmeter, and rheostat in electrical circuits.<br>Explain the use of a tester and a multimeter.                                                                                                                                                                                                                       | P | CO1 | Understand | 2 |
| Voltage and current measurement             | Demonstrate the measurement of voltage and current in a DC circuit.<br>1. Select an appropriate instrument and range for measuring DC voltage/current.<br>2. Demonstrate correct meter reading technique<br>3. Compute the actual reading using the correct multiplication factor (Use a multi-range meter and apply the multiplication factor to obtain the correct value) | P | CO1 | Apply      | 6 |
| Module II                                   |                                                                                                                                                                                                                                                                                                                                                                             |   |     |            |   |
| Basics of AC systems                        | Explain basic terms in AC systems such as cycle, frequency, time period, angular frequency, amplitude, instantaneous value, maximum value, rms value, average value, form factor, and peak factor. State their SI units.<br>Write the instantaneous equation of the AC voltage and current.                                                                                 | L | CO2 | Understand | 3 |
| Electrical Safety                           | Describe standard electrical safety precautions to be followed in electrical installations.                                                                                                                                                                                                                                                                                 | L | CO2 | Understand | 1 |
| Electrical Safety                           | Identify the safety devices used in Electrical systems (fuse, MCB, ELCB, RCCB).<br>State the functions of safety devices used in electrical systems, including fuses, miniature circuit breakers (MCB), earth leakage circuit breakers (ELCB), and RCCB.                                                                                                                    | L | CO2 | Remember   | 2 |
| AC Power                                    | Define different types of power in AC circuits(active power, reactive power, apparent power).<br>Define the terms phase angle and power factor.                                                                                                                                                                                                                             | L | CO2 | Remember   | 2 |
| AC Power                                    | Solve numerical problems related to power calculations in single-phase AC circuits.                                                                                                                                                                                                                                                                                         | L | CO2 | Apply      | 3 |
| Electrical Safety and Safety devices        | Discuss electrical safety precautions.                                                                                                                                                                                                                                                                                                                                      | P | CO2 | Understand | 3 |
| Electrical Safety and Safety devices        | Summarize the electrical safety devices in residential installations. (Fuse, ELCB, MCCB, RCCB)                                                                                                                                                                                                                                                                              | P | CO2 | Understand | 3 |
| Measurement of AC parameters                | Demonstrate a simple ac circuit containing a source, switch, fuse, and a load and measure voltage, current and power.                                                                                                                                                                                                                                                       | P | CO2 | Apply      | 3 |
|                                             | Practical Series Exam - I                                                                                                                                                                                                                                                                                                                                                   | P | CO2 | Apply      | 3 |
| Module III                                  |                                                                                                                                                                                                                                                                                                                                                                             |   |     |            |   |
| Passive Components                          | Define resistance and state its units.<br>Describe types of resistors: fixed (carbon composition, wire-wound) and variable resistors(Potentiometer, Rheostat, Trimmer). Summarize various applications of resistors in electronic circuits.                                                                                                                                 | L | CO3 | Understand | 2 |
| Passive Components                          | Write the value of a resistor using the standard four-band color-coding method.                                                                                                                                                                                                                                                                                             | L | CO3 | Remember   | 2 |
| Passive Components                          | Define inductance, self-inductance, and mutual inductance and state their units.<br>List the various applications of inductors.                                                                                                                                                                                                                                             | L | CO3 | Remember   | 2 |
| Passive Components                          | Define capacitance,<br>Classify types of capacitors-fixed(Ceramic, film type, mica and electrolytic), variable(tuning capacitor, trimmer capacitor),, and their applications,<br>Describe series and parallel combinations of capacitors (equations only).                                                                                                                  | L | CO3 | Understand | 2 |
| Problems on series and parallel combination | Solve numerical problems related to series and parallel combinations of capacitance.                                                                                                                                                                                                                                                                                        | L | CO3 | Apply      | 3 |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |     |            |   |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|------------|---|
| Electronic circuit                        | Identify breadboard connection points.<br>Construct a circuit connecting a 3V LED to 5V battery via a switch and resistor on a breadboard.                                                                                                                                                                                                                                                                                                         | P | CO3 | Apply      | 3 |
| DSO                                       | Operate a DSO<br>1. Identify oscilloscope voltage and time scale controls.<br>2. Recognize and select the correct probe setting (1× or 10×) on the scope and probe.<br>3. Demonstrate proper probe checking/compensation using the calibration signal.<br>4. Apply voltage/div and time/div to obtain a scaled waveform.<br>5. Compute maximum voltage (Vmax), peak-to-peak voltage (Vpp), and average voltage (Vavg) from the displayed waveform. | P | CO3 | Apply      | 3 |
| Measurement of resistance and capacitance | Determine the value of a given resistor using the color coding method and verify it using a digital multimeter. Identify common capacitor types (mica, ceramic, electrolytic)<br>Verify the capacitance values using manufacturer codes and an LCR meter.                                                                                                                                                                                          | P | CO3 | Apply      | 3 |
| Measurement of resistance and capacitance | Construct an inductor by winding insulated copper wire around a toroidal/EE core and measure the inductance value using an LCR meter(no design required).                                                                                                                                                                                                                                                                                          | P | CO3 | Apply      | 3 |
| Module IV                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |     |            |   |
| Active & Passive components               | Summarize active and passive components                                                                                                                                                                                                                                                                                                                                                                                                            | L | CO4 | Understand | 1 |
| Types of semiconductors                   | Summarize types of semiconductors-Extrinsic, Intrinsic- N type, P type, and doping                                                                                                                                                                                                                                                                                                                                                                 | L | CO4 | Understand | 2 |
| Diode                                     | Explain the construction, working, and biasing of pn junction diodes                                                                                                                                                                                                                                                                                                                                                                               | L | CO4 | Understand | 3 |
| Rectifier                                 | Explain half wave rectifier circuit                                                                                                                                                                                                                                                                                                                                                                                                                | L | CO4 | Understand | 2 |
| Transistors                               | Explain the construction and working of transistors, types- NPN and PNP                                                                                                                                                                                                                                                                                                                                                                            | L | CO4 | Understand | 2 |
| Logic gates                               | Summarize the truth table of Logic gates- AND, OR, NOT, NAND, NOR, XOR, XNOR.                                                                                                                                                                                                                                                                                                                                                                      | L | CO4 | Understand | 2 |
| Diode and transistor                      | Identify the terminals of diode and transistor                                                                                                                                                                                                                                                                                                                                                                                                     | P | CO4 | Remember   | 3 |
| Half wave rectifier                       | Construct a half wave rectifier with and without filter and plot the input / output waveforms                                                                                                                                                                                                                                                                                                                                                      | P | CO4 | Apply      | 3 |
| Truth table verification                  | Demonstrate the truth table verification of various logic gates AND, OR, NOT, NAND, NOR, XOR                                                                                                                                                                                                                                                                                                                                                       | P | CO4 | Apply      | 3 |
|                                           | Practical Series Exam - II                                                                                                                                                                                                                                                                                                                                                                                                                         |   | CO4 | Apply      | 3 |

#### Suggested Student Activities for Self-Learning

| Week      | Self-Learning description                                                                                                                                                                                                                                                                                   | Hours |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Module I  |                                                                                                                                                                                                                                                                                                             |       |
| 1         | List the rated voltage and power printed on the common household appliances (fan, light, iron, mixer, LED bulb, mobile charger, fridge, washing machine, TV, etc.)                                                                                                                                          | 6     |
| 2         | Tabulate the output voltage (V) and output current (A) ratings found on the labels of three different charging adapters (e.g., a phone charger, laptop charger, and router adapter).                                                                                                                        | 6     |
| 3         | Solve five simple practice problems to find the circuit parameters, such as voltage, current, and resistance in a circuit using Ohm's Law                                                                                                                                                                   | 6     |
| 4         | Construct a circuit in an online simulator (e.g., Tinkercad) with a battery and two resistors. Connect virtual meters to prove that voltage divides in a series circuit, but current remains the same.                                                                                                      | 6     |
| Module II |                                                                                                                                                                                                                                                                                                             |       |
| 5         | Using the standard Indian household AC supply (230 V RMS, 50 Hz), sketch the instantaneous voltage equation $v(t) = V_m \sin(\omega t)$ for one complete cycle. Calculate $V_m$ , $V_{rms}$ , $V_{avg}$ , frequency $f$ , and time period $T$ , then label these values clearly on the sinusoidal waveform. | 6     |
| 6         | Summarize the effects of electric shocks and its precautions.                                                                                                                                                                                                                                               | 3     |
| 7         | Identify the safety devices installed in the main distribution board (DB) at home. Record the current ratings (in amperes) and the earth-leakage sensitivity of the RCCB/ELCB (in mA). Prepare a one-page note explaining the difference between an MCB trip and an RCCB/ELCB trip                          | 3     |

|                           |                                                                                                                                                                                                                                            |    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 8                         | Calculate the total connected load of your house wiring using the ratings of appliances and fixtures.                                                                                                                                      | 12 |
| Module III                |                                                                                                                                                                                                                                            |    |
| 9,10                      | Watch a video on capacitor code decoding and prepare a PowerPoint that outlines the step-by-step procedure to determine capacitance values from manufacturer codes for classroom presentation.                                             | 12 |
| 11                        | Collect capacitors of different types (such as ceramic, electrolytic, film, and tantalum) and identify and record their specifications including capacitance value, voltage rating, tolerance, and polarity, and compare your observations | 6  |
| 12                        | Construct an inductor by winding insulated copper wire around a toroidal/EE core (no design calculations required) using a video tutorial and measure the inductance value using a LCR meter                                               | 6  |
| Module IV                 |                                                                                                                                                                                                                                            |    |
| 13                        | Refer to the datasheet of a PN junction diode(available) and prepare a detailed list of its key specifications including electrical, thermal, and mechanical parameters                                                                    | 6  |
| 14                        | Identify the components used in a mobile charger                                                                                                                                                                                           | 6  |
| 15                        | Construct a basic hobby circuit using a breadboard, resistors, LEDs, and a battery pack                                                                                                                                                    | 6  |
| Total Self-Learning hours |                                                                                                                                                                                                                                            | 90 |

### Learning Resources

| TEXT BOOK |                                                                                     |                                               |                                  |
|-----------|-------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------|
| Sl. No.   | Title                                                                               | Author                                        | Publication                      |
| 1         | A text book of Electrical Technology Part1"Basic Electrical Engineering in SI units | B. L. Thereja and A. K. Thereja,              | S- Chand Publications 2012       |
| 2         | Basic Electronics and Linear Circuits                                               | N.N. Bhargava , D.C. Kulshreshtha S.C. Gupta, | Tata McGraw Hill Education, 1983 |
| 3         | Basic Electrical Engineering                                                        | V.K. Mehta & Rohit Mehta                      | S. Chand & Company Ltd.          |

| REFERENCE |                                 |                             |                            |
|-----------|---------------------------------|-----------------------------|----------------------------|
| Sl. No.   | Title                           | Author                      | Publication                |
| 1         | Basic Electrical Engineering    | D.P. Kothari & I.J. Nagrath | Tata McGraw-Hill Education |
| 2         | Electronic Devices and Circuits | G K Mithal                  | Khanna Publishers, 1988    |
| 3         | Basic Electrical Engineering    | M.S. Naidu & S. Kamakshaiah | Tata McGraw-Hill Education |

| WEB RESOURCES |                                                                                                                                           |                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Sl. No.       | URL                                                                                                                                       | Modules Covered |
| 1             | <a href="http://www.electical4u.com">www.electical4u.com</a>                                                                              | All             |
| 2             | <a href="https://www.digimat.in/nptel/courses/video/108105112/L01.html">https://www.digimat.in/nptel/courses/video/108105112/L01.html</a> | All             |
| 3             | <a href="https://www.digimat.in/nptel/courses/video/108101091/L01.html">https://www.digimat.in/nptel/courses/video/108101091/L01.html</a> | All             |
| 4             | Coursera - Electrical Engineering Courses                                                                                                 | All             |

### Major Equipment/Instruments/Components required for a batch of 30 students

| Sl. No. | Particulars                 | Specifications                           | Quantity |
|---------|-----------------------------|------------------------------------------|----------|
| 1       | DSO                         | 25MHz-50MHz, 2 channel                   | 6        |
| 2       | Function generator          | 3MHz, Sine/Square/Triangular wave output | 6        |
| 3       | Dc regulated power supplies | 0-30V, 2A (dual channel preferred)       | 6        |
| 4       | Wattmeter                   | LPF or UPF, 250V, 5A                     | 6        |
| 5       | Digital Multimeter          | Standard handheld                        | 6        |
| 6       | LCR meter                   | Handheld or benchtop                     | 6        |
| 7       | Auto-Transformer            | 1 phase, 230V input, 0-270V output, 4A   | 6        |
| 8       | Digital IC trainer kit      | Built in 5V supply, breadboard, clock    | 6        |
| 9       | Rheostats                   | wire wound, 500hms, 5A                   | 6        |
| 10      | DC voltmeter and ammeter    | Analog (moving coil), 0-30V and 0-100mA  | 6 each   |

|    |                           |                                                                                                 |                             |
|----|---------------------------|-------------------------------------------------------------------------------------------------|-----------------------------|
| 11 | AC voltmeter and ammeters | Analog (moving iron), 0-300V and 0-5A                                                           | 6 each                      |
| 12 | Logic ICs                 | 7408 (AND), 7432 (OR), 7404 (NOT), 7400 (NAND), 7402 (NOR), 7486 (XOR)                          | 10 of each type             |
| 13 | Active Components         | Diodes (1N4007), Transistors (BC547 NPN), LEDs (Red/Green)                                      | 20 of each                  |
| 14 | Passive Components        | Resistors (Assorted values: 1kΩ, 10kΩ, 330Ω), Capacitors (10μF, 100μF), Inductors (small value) | 1 Box (Assorted)            |
| 15 | Prototyping               | Breadboards (800 tie points), Connecting wires (Single strand 23/24 SWG)                        | 6 Boards + 2 Spools of wire |
| 16 | Transformers              | Step-down transformer (230V to 9V or 12V)                                                       | 6                           |

#### Major Software required for a batch of 30 students

| Sl. No. | Particulars | Specifications   | Quantity |
|---------|-------------|------------------|----------|
| 1       | Tinkercad   | Free (web based) |          |
| 2       | LTspice     | Free             |          |
| 3       | Logisim     | Free (web based) |          |

#### Assessment Methodology - Practicum

| Mode               | Attendance      | CA1                                                  | CE                    | CA2                                        | CA3                                         | CA4                     | CA5                     | ESE                   |                            |
|--------------------|-----------------|------------------------------------------------------|-----------------------|--------------------------------------------|---------------------------------------------|-------------------------|-------------------------|-----------------------|----------------------------|
|                    |                 | Average of Self learning activities from each module | Continuous evaluation | Practical test (first half of experiments) | Practical test (second half of experiments) | Series Exam (2 modules) | Series Exam (2 modules) | Written Exam (theory) | Lab Examination (internal) |
| Duration           |                 |                                                      |                       | 3 hours                                    | 3 hours                                     | 2 hours                 | 2 hours                 | 2.5 hours             | 3 hours                    |
| Exam mark          |                 | 15                                                   | 50                    | 40                                         | 40                                          | 50                      | 50                      | 60                    | 40                         |
| Converted to       |                 | 5                                                    | 10                    | 10                                         | 10                                          | 10                      | 10                      |                       |                            |
| Marks              | 5               | 5                                                    | 10                    | 10                                         |                                             | 10                      |                         | 60                    | 40                         |
| Total              | 40              |                                                      |                       |                                            |                                             |                         |                         | 100                   |                            |
| Tentative schedule | End of semester | By 4th week                                          | By 7th week           | By 11th week                               | By 14th week                                | 8th week                | 15th week               | End of semester       |                            |

#### Scheme of Evaluation - Practical (Continuous Internal Evaluation)

| Sl. No. | Criteria                           | Description                                                                                                                  | Marks |
|---------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------|
| 1       | Preparation & Punctuality          | Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.             | 10    |
| 2       | Experimental Setup & Procedure     | Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure. | 10    |
| 3       | Observation & Data Recording       | Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.                    | 5     |
| 4       | Analysis & Result Interpretation   | Proper calculations, graphical representation, result verification, discussion, and logical conclusion.                      | 10    |
| 5       | Viva Voce / Concept Understanding  | Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.            | 10    |
| 6       | Workmanship, Discipline & Teamwork | Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.                                        | 5     |
| Total   |                                    |                                                                                                                              | 50    |

#### Scheme of Evaluation - Practical Test

| Category                                 | Things to evaluate                                                                                           | Marks |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------|
| Write-up / Procedure (before experiment) | Aim, circuit diagram/flowchart, stepwise procedure.                                                          | 10    |
| Experiment Setup & Execution             | Correct circuit connections / coding / setup.<br>Handling of instruments/software.<br>Accuracy of execution. | 10    |

|                               |                                                                                                                    |    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|----|
| Observation & Result / Output | Proper tabulation of readings.<br>Calculations/graphs/program output.<br>Correctness of result / expected outcome. | 8  |
| Viva Voce                     | Conceptual understanding of experiment.<br>Related theoretical knowledge.                                          | 8  |
| Record                        | Completion & neatness of practical record.<br>Proper certification by faculty.                                     | 4  |
| Total                         |                                                                                                                    | 40 |

#### Series Examination Pattern- Theory

| Time    | Module         | Part A | Marks | Part B | Marks | Part C<br>(With choice) | Marks | Total |
|---------|----------------|--------|-------|--------|-------|-------------------------|-------|-------|
| 2 hours | Any one module | 2      | 1     | 3      | 3     | 2                       | 7     | 25    |
|         | Any one module | 2      | 1     | 3      | 3     | 2                       | 7     | 25    |
|         | Number         | 4      |       | 6      |       | 4                       |       | 50    |

#### End Semester Examination Pattern of Question Paper - Theory

| Duration  | Module | TYPE OF QUESTIONS                        |                        |                                                  |                         |                                                          |                         | Total Marks |
|-----------|--------|------------------------------------------|------------------------|--------------------------------------------------|-------------------------|----------------------------------------------------------|-------------------------|-------------|
|           |        | Part A<br>(2 Questions from each module) |                        | Part B<br>2 out of 3 questions from each module) |                         | Part C<br>(1 set questions with choice from each module) |                         |             |
|           |        | No. of Questions                         | Marks<br>(1 mark each) | No. of Questions                                 | Marks<br>(3 marks each) | No. of Questions                                         | Marks<br>(7 marks each) |             |
| 2.5 hours | 1      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 2      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 3      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 4      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           |        | 8                                        | 8                      | 12                                               | 24                      | 4 sets                                                   | 28                      | 60          |